# Lewis Scott Cullinan

MSc Robotics student | First-Class BEng Electrical & Electronic Engineering

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## PROFILE

Electrical and electronic engineering graduate now studying robotics. My work connects embedded control, sensing and software, with a growing focus on autonomous systems that can operate reliably in uncertain environments.

## EDUCATION

**MSc Robotics** — current study

**Liverpool John Moores University** | BEng (Hons) Electrical & Electronic Engineering, 2022–2025 | **First-Class Honours**

Selected marks: Automation & IoT 95%; self-guided solar smart irrigation project 87%; Advanced Mathematics 100%.

**Riverside College** | C&G Electrical Installations Level 3, 2016–2018. **Parklands High School** | 12 GCSEs, 2008–2013.

## SELECTED PROJECTS

- **SmartWater — solar smart irrigation and IoT.** Arduino-based soil monitoring and automated watering with a local display and web dashboard. The degree project combined sensor calibration, pump control, ESP8266 connectivity and environmental monitoring; awarded 87%.
- **LEO rover — autonomous block sorting.** Four-wheel mobile platform for search, block detection, retrieval and sorting. Project development brings navigation, object state and manipulation together; performance figures are not claimed here.
- **Line follower.** Mobile robot project exploring calibrated sensing, steering feedback and recovery when a marked route is lost.
- **Wind turbine drone inspection.** Engineering design for aerial image capture, coverage tracking and defect review. The inspection workflow is a design study, with no claimed turbine flight results.

## ADDITIONAL ENGINEERING WORK

- **FPGA washing-machine controller:** VHDL finite-state machine on a Basys 3 Artix-7 board using Xilinx Vivado; timing transitions, double-wash cycle and seven-segment output.
- **Signal processing:** MATLAB and Simulink work with Fourier transforms, filtering, convolution and control-related analysis.

## TECHNICAL SKILLS

**Robotics:** ROS 2, SLAM and localisation, Nav2, MoveIt 2, Gazebo, RViz, computer vision and simulation workflows.

**Programming:** Python, C/C++, C#, Assembly, VHDL; basic Verilog; HTML, CSS and JavaScript for IoT dashboards.

**Electronics:** Arduino, ESP8266, STM32 and FPGA platforms; sensor integration, motor and pump control, PCB design, soldering and prototype testing.

**Tools:** Arduino IDE, Xilinx Vivado, MATLAB/Simulink, KiCad, Multisim, Proteus, Quartus and Git/GitHub.

## EXPERIENCE

**Royal Liverpool Hospital — Electrician's Mate (work experience), 2017:** Assisted qualified electricians with wiring, lighting and socket installation, inspection and testing.

**Marks & Spencer and B&Q, 2015–2020:** Stock handling, logistics and customer service, developing teamwork and problem-solving under time pressure.

## ADDITIONAL INTERESTS

Piano and other instruments, reading, fitness, drone flying and electric unicycles.